of a learning curve because so many different options are available, but the Ecto.Query documentation[20] is very comprehensive.

And with this, our SmsMessageStore is complete! We could stop here and use the store module in our application, but there's a code organization technique that helps create clarity. Let's create a context to wrap our functions.

Put Everything in a Context

Code organization has been and always will be a hot topic. People have many different opinions. Whether right or wrong, these opinions tend to solidify into best practices based on the community's preference. A code organization technique called "contexts" has taken hold in the Elixir community.

Let's break down what contexts are and then create one of our own.

What Is a Context?

Although they can be intimidating at first, contexts are simple. They are modules that group together related functions. This grouping is typically done based on the domain of the application you're building—which means you get to decide the context boundaries.

Contexts define a public API between different parts of your application. This makes it harder for spaghetti code to pop up in your app because the callable functions are clearly available in the context module. This context module is typically one level below your main application module prefix.

In our PhoneApp application, we'll create a context module called PhoneApp.Conversations. This module holds all of the functions that can be used by other parts of the application (like Phoenix Controllers, which we'll cover in the next chapter.)

There's nothing stopping other parts of the application from reaching in and "violating" the context boundary. It's more like an agreement to not do so.

The biggest obstacle for developers starting out with contexts is "what should I put in my context module?" The best answer is to make a decision with the information you have on hand and don't sweat the details too much. You can always change the context as needed. And you don't need to strictly follow the context rules—nothing prevents you from making the best decision for your app.

20. https://hexdocs.pm/ecto/Ecto.Query.html#content